USN | | | | | | | | | | [22MTAIDE61]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER - 2024

GAME THEORY

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Compare and contrast cooperative and non-cooperative games. How do the potential for binding agreements influence player strategies and outcomes? Provide detailed examples of scenarios where each type is applicable.	**CO1**	**L4**	**10**
	b.	Demonstrate the simple card game using normal representation and strategic form.	**CO1**	**L3**	**10**
		OR			
2.	**a.**	Provide a comprehensive analysis of the Prisoner's Dilemma with ordinal preferences, prompting exploration of its structure, equilibria, and significance in strategic interactions.	**CO2**	**L2**	**10**
	b.	Examine the concepts of mixed strategies and correlated equilibrium in game theory.	**CO2**	**L4**	**10**
3.	**a.**	The political figures Rosa and Ernesto each has to take a position on an issue. The options are Berlin (B) or Havana (H). They choose sequentially. A third person, Karl, determines who chooses first. Both Rosa and Ernesto care only about the actions they choose, not about who chooses first. Rosa prefers the outcome in which both she and Ernesto choose B to that in which they both choose H, and prefers this outcome to either of the ones in which she and Ernesto choose different actions; she is indifferent between these last two out comes. Ernesto's preferences differ from Rosa's in that the roles of B and H are reversed. Karl's preferences are the same as Ernesto's. Model this situation as an extensive game with perfect information. Specify the components of the game and represent the game in a diagram.	**CO2**	**4**	**10**
	b.	Find all the subgames of the game in 3a.	**CO2**	**4**	**10**
		OR			
4.	**a.**	Describe a subgame perfect equilibrium, and how is it related to sequential rationality? Why is it important for ensuring rationality at every point in the game?	**CO3**	**4**	**10**
	b.	What is the folk theorem in the context of repeated games? Summarize its main implications for players' strategies and outcomes.	**CO3**	**2**	**10**

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5.	**a.**	Consider the set of players is N = {I ,2,3} and the set of strategy options for each player i is $C_i = \{(x_1, x_2, x_3) \in \mathbf{R}^3 \mid x_1 + x_2 + x_3 \leq 300 \text{ and, } \forall j,\ x_j \geq 0\}$. the players get 0 unless all three players propose the same allocation. Describe the characteristic functions, minimax representation in coalitional form.	**CO4**	**2**	**10**
	b.	What is a coalition in the context of cooperative games? How do players form coalitions, and what factors influence their decisions to join or leave?	**CO4**	**2**	**10**
		OR			
6.	**a.**	What is the Nash bargaining solution, and what are its key properties that distinguish it from other negotiation models.	**CO4**	**2**	**10**
	b.	What are the key properties of the Shapley value, and how do they contribute to its interpretation as a fair distribution of payoffs among players in a cooperative game?	**CO4**	**2**	**10**
7.	**a.**	How does social network analysis contribute to our understanding of individual behavior and group dynamics in social contexts.	**CO5**	**2**	**10**
	b.	In what ways can network games be applied to analyze behaviors in social media platforms or online communities?	**CO5**	**2**	**10**
		OR			
8.	**a.**	How does the concept of Nash equilibrium apply to voting scenarios, and what does it reveal about voter behavior in competitive elections.	**CO5**	**2**	**10**
	b.	Evaluate the real-world applications of any two game theory in economics and politics.	**CO5**	**2**	**10**